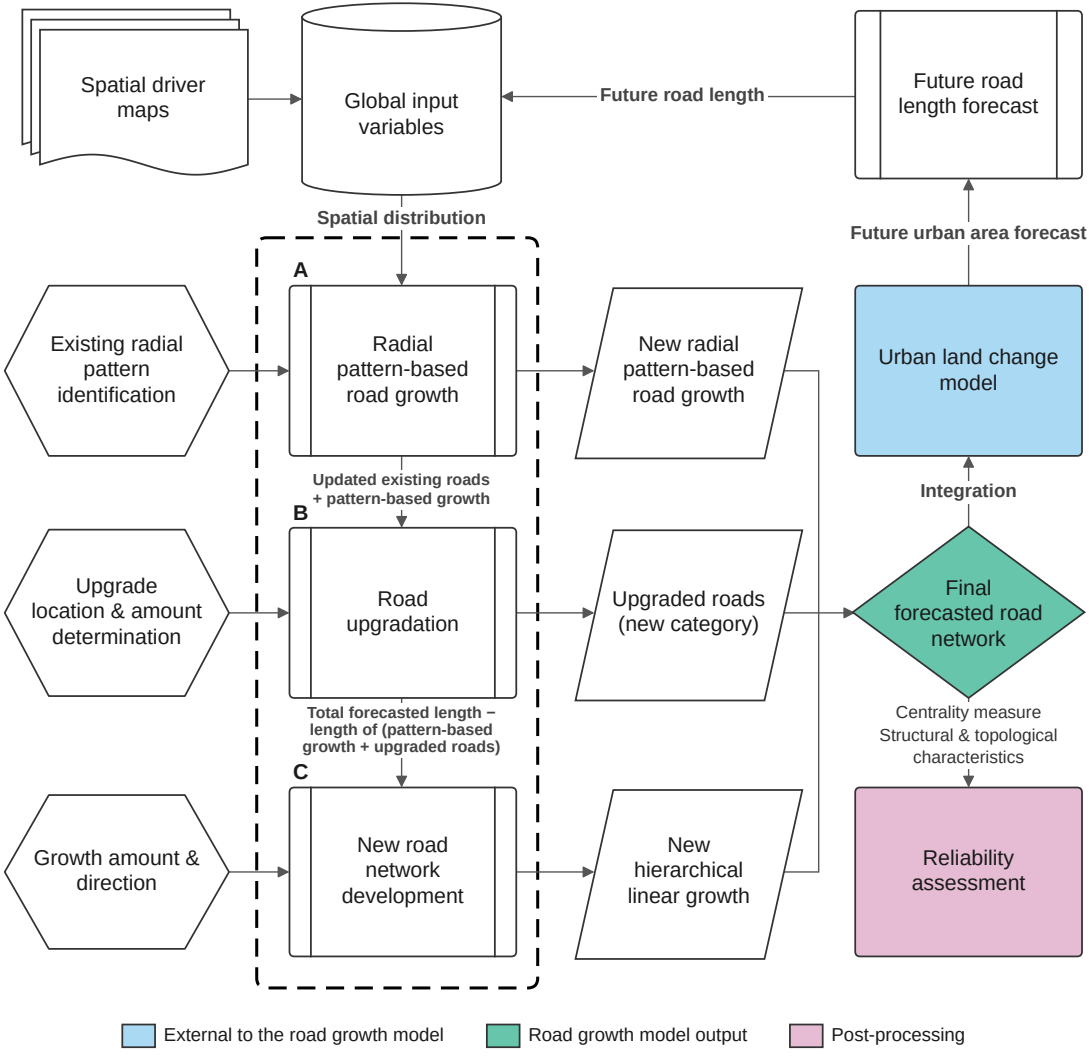
Spatial driver maps
Global input variables
Future road length
Future road length forecast
Spatial distribution
Future urban area forecast
A
Existing radial pattern identification
Radial pattern-based road growth
New radial pattern-based road growth
Urban land change model
Updated existing roads + pattern-based growth
Integration
B
Upgrade location & amount determination
Road upgradation
Upgraded roads (new category)
Final forecasted road network
Total forecasted length − length of (pattern-based growth + upgraded roads)
Centrality measure
Structural & topological characteristics
C
Growth amount & direction
New road network development
New hierarchical linear growth
Reliability assessment
External to the road growth model
Road growth model output
Post-processing